

Task 3 – Lecture Slide Generation Using LLMs

Student Information

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Lecture Slide Generation Using LLMs – Report

Objective

The objective of this task was to use Large Language Models (LLMs) to analyze the Orientation Lecture (Lecture 01: Introduction to General Agent Problem Solving - GAPS) and generate a professional presentation covering the lecture topics.

AI Tools and Models Used

1. Google Gemini

Used for:

- Understanding the YouTube lecture
- Creating a detailed lecture summary
- Extracting key concepts
- Generating the final presentation slides

2. ChatGPT

Used for:

- Refining the lecture summary
- Designing the presentation structure
- Creating a detailed 20-slide outline
- Improving content organization

3. Gamma (Attempted)

Used initially for automatic slide generation. However, the free version only generated approximately 10 slides, which was insufficient for the assignment requirements.

Prompt Iteration 1

Prompt

Watch this lecture and create:

1. A detailed summary
2. Key topics discussed
3. Important concepts
4. Timeline of topics covered
5. Suggested presentation structure (15–20 slides)

Video: <https://www.youtube.com/watch?v=a1NIVLrComg>

Result

Gemini successfully:

- Understood the lecture
- Extracted major concepts
- Produced a structured summary
- Suggested a logical presentation flow

What Worked Well

- Generated detailed lecture notes
- Identified the major themes accurately
- Saved significant manual effort

Prompt Iteration 2

Prompt

Create a professional 20-slide presentation from this lecture summary.

Requirements:

- Student audience
- Include slide titles
- 4–6 bullet points per slide
- Include speaker notes
- Include learning objectives
- Include conclusion and key takeaways

Result

ChatGPT transformed the summary into a structured presentation outline.

What Worked Well

- Produced a clear slide sequence
- Added learning objectives
- Maintained logical flow
- Included presentation best practices

Improvement Over Previous Prompt

Instead of only summarizing the lecture, the AI was guided to create presentation-ready content.

Prompt Iteration 3

Prompt

Convert this lecture summary into a 20-slide presentation.

For each slide provide:

- Title
- 4–6 bullet points
- Speaker notes

Result

Gemini generated the final presentation content and produced the slide deck.

What Worked Well

- Generated complete slide content
 - Included speaker notes
 - Produced presentation-ready material
 - Required minimal manual editing
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What Did Not Work Well

Gamma Free Version

Issue:

- Generated only about 10 slides.

- Reduced content depth.
- Could not fully satisfy the 20-slide requirement.

Solution:

- Switched to a combination of Gemini and ChatGPT.
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How the Prompts Were Refined

The prompts gradually became more specific.

Initial Prompt

Focused only on understanding the lecture.

Refined Prompt

Specified:

- Audience
- Number of slides
- Slide structure
- Speaker notes
- Learning objectives
- Key takeaways

Adding these requirements significantly improved the quality and usefulness of the output.

Complete Process Followed

Step 1

Provided the YouTube lecture URL to Gemini.

Step 2

Generated:

- Lecture summary
- Key concepts
- Topic timeline
- Suggested slide structure

Step 3

Provided the summary to ChatGPT.

Step 4

Requested a professional 20-slide presentation outline with speaker notes.

Step 5

Provided the refined content back to Gemini.

Step 6

Used Gemini to generate the final presentation slides.

Step 7

Reviewed and finalized the presentation.

Key Learnings

- AI can quickly transform long-form lectures into structured learning material.
 - Prompt quality directly affects output quality.
 - Providing clear requirements significantly improves results.
 - Combining multiple AI tools often produces better outcomes than using a single tool.
 - Iterative prompting is an effective method for generating professional presentations.
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Conclusion

This task demonstrated how Large Language Models can be used to analyze educational content, summarize complex topics, and generate professional presentations. By combining Gemini and ChatGPT and refining prompts iteratively, a complete presentation on "Introduction to General Agent Problem Solving (GAPS)" was successfully created.